Name _____ Grade _____

ME 5130 Advanced Heat and Mass Transfer, Fall 2026

Midterm Exam – Take-Home portion
Two questions, 60 points

Assigned: Wednesday, October 21, 2026, 6:15 pm ET
Due: Wednesday, October 28, 2026, 6:15 pm ET

2. Plotting of the Temperature Solution (2 parts: 2a to 2b)

Consider the assigned homework problem 2-15 from Ozisik (1993). Use a = 1 m, b = 1 m, k = 0.8 W/m-K, h = 10 W/m^2-K, and T = f(x) = 100 K.

2a. List the first ten eigenvalues β_m used to generate your contour plot. Attach a copy of your code.

2b. Create a 2-D contour plot of the temperature field T(x,y) for [0 ≤ x ≤ a; 0 ≤ y ≤ b]. Show at least 6 isotherms, and numerically label the temperature value of all the isotherms. Attach a copy of your code.

3. Diffusion Mass Transfer (5 parts: 3a to 3e)

For this problem, consider example problem 17.4-1 in Bird, Stewart and Lightfoot (1960), pg. 534-535.

3a. Using the Shell Mass Balance approach:

$$[\text{in}] - [\text{out}] + [\text{reaction}] = 0$$

create a sketch of a control volume, and indicate all the in/out fluxes, and reaction terms in your sketch.

3b. Use this Shell Mass Balance sketch to derive the following governing equation and boundary conditions.

Governing equation: $$-\mathfrak{D}_{AB}\frac{\partial^2 c_A}{\partial z^2} + k_1''' c_A = 0$$

Boundary conditions: $$c_A = c_{A0} \;\; at \;\; z = 0$$
$$c_A = c_{A\delta} \;\; at \;\; z = \delta$$

All assumptions must be clearly stated, and all steps clearly explained.

3c. Show that the solution to this system is the following. All assumptions must be clearly stated, and all steps clearly explained.

$$\frac{c_A}{c_{A0}} = \frac{\Gamma \sinh b_1\zeta + \sinh b_1(1-\zeta)}{\sinh b_1}$$

3d. Now, assume that there is no chemical reaction. Since $k_1''' = 0$, the governing equation is now:

$$-\mathfrak{D}_{AB}\frac{d^2c_A}{dz^2} = 0$$

and subjected to the same boundary conditions. Solve the differential equation, and show that the solution without chemical reaction is:

$$\left(\frac{c_A}{c_{A0}}\right)_{no\ rxn} = \Gamma\zeta + (1-\zeta)$$

Do not just set $b_1 = 0$ in the solution given in problem 2c. All assumptions must be clearly stated, and all steps clearly explained.

3e. Using Fick's law, show that the fluxes with and without reaction at z = 0 are:

With chemical reaction: $(N_{Az})_{z=0} = \frac{\mathfrak{D}_{AB}c_{A0}}{\delta}\left(\frac{b_1\cosh b_1 - b_1\Gamma}{\sinh b_1}\right)$

No chemical reaction: $(N_{Az})_{z=0} = \frac{\mathfrak{D}_{AB}c_{A0}}{\delta}(1-\Gamma)$